

Lecture 16: Radial Basis Functions

Caltech CS156 Technical Study Guide

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Executive Summary

Radial Basis Functions (RBF) construct hypotheses by combining localized distance-based bell curves. RBF bridges instance-based learning (Nearest Neighbors), neural networks, kernel methods, and regularization theory. This guide covers K -means clustering (Lloyd's Algorithm) for unsupervised center selection, pseudo-inverse weight fitting, and comparisons with SVMs.

1 The RBF Model Formulation

An RBF network evaluates the Euclidean distance $\|\mathbf{x} - \boldsymbol{\mu}_k\|$ between input $\mathbf{x}$ and K center vectors $\boldsymbol{\mu}_k$:

$$h(\mathbf{x}) = \sum_{k=1}^K w_k \exp(-\gamma \|\mathbf{x} - \boldsymbol{\mu}_k\|^2) + b \quad (1)$$

2 Exact RBF vs. K -Centers RBF

- **Exact Interpolation** ($K = N$): Centers are fixed at every training point ($\boldsymbol{\mu}_n = \mathbf{x}_n$). $E_{\text{in}} = 0$, but high risk of overfitting when noise is present.
- **K -Centers RBF** ($K \ll N$): Selects K representative cluster centers $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K$ to capture data density without overfitting.

3 Unsupervised Center Selection: K -Means (Lloyd's Algorithm)

Centers $\boldsymbol{\mu}_k$ are determined without labels y_n by minimizing intra-cluster variance:

$$\min_{\mathbf{S}, \boldsymbol{\mu}} \sum_{k=1}^K \sum_{\mathbf{x}_n \in S_k} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2 \quad (2)$$

3.1 Lloyd's Algorithm Steps

1. **Initialization:** Select K initial centers $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K$ randomly.
2. **Assignment Step:** Assign each point $\mathbf{x}_n$ to the nearest center:

$$S_k = \{\mathbf{x}_n : \|\mathbf{x}_n - \boldsymbol{\mu}_k\| \leq \|\mathbf{x}_n - \boldsymbol{\mu}_j\| \quad \forall j\} \quad (3)$$

3. **Update Step:** Recalculate cluster centroids:

$$\boldsymbol{\mu}_k = \frac{1}{|S_k|} \sum_{\mathbf{x}_n \in S_k} \mathbf{x}_n \quad (4)$$

4. **Convergence:** Repeat assignment and update steps until centers stabilize.

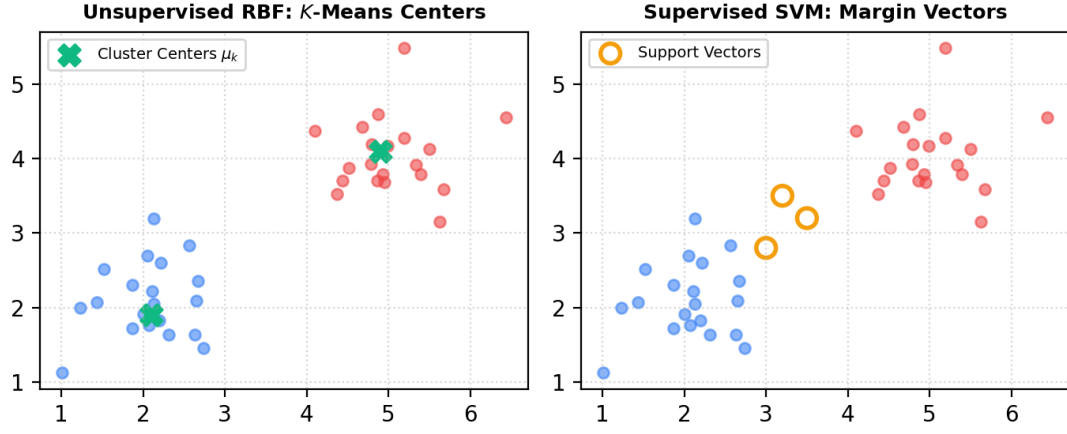


Figure 1: RBF Centers (K -Means Unsupervised) vs. SVM Support Vectors (Supervised Margin Maxima).

4 Supervised Weight Optimization

Once centers μ_k are fixed, the RBF model is linear in weights $\mathbf{w}$. Construct the design matrix $\Phi \in \mathbb{R}^{N \times K}$:

$$\Phi_{nk} = \exp(-\gamma \|\mathbf{x}_n - \mu_k\|^2) \quad (5)$$

Solve for weights $\mathbf{w}$ using the Moore-Penrose Pseudo-Inverse:

$$\mathbf{w} = \Phi^\dagger \mathbf{y} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y} \quad (6)$$